
Problem Set 7

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PY 451 Section A1

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1 Practice with Kets and Wavefunctions

1.

$$\begin{aligned}\langle \psi | \psi \rangle &= \int_{\mathbb{R}} |\psi(x)|^2 dx \\ &= \int_{-\infty}^{\infty} |A|^2 \left| e^{-a|x-b|+ivx} \right|^2 dx \\ &= |A|^2 \int_{-\infty}^{\infty} \left| e^{-a|x-b|} e^{ivx} \right|^2 dx \\ &= |A|^2 \int_{-\infty}^{\infty} \left| e^{-a|x-b|} \right|^2 |e^{ivx}|^2 dx \\ &= |A|^2 \int_{-\infty}^{\infty} \left(e^{-a|x-b|} \right)^2 1^2 dx \\ &= |A|^2 \left(\int_{-\infty}^b e^{2a(x-b)} dx + \int_b^{\infty} e^{-2a(x-b)} dx \right).\end{aligned}$$

This last step comes from the fact that

$$|x-b| = \begin{cases} x-b, & x-b \geq 0 \\ -(x-b), & x-b \leq 0 \end{cases} = \begin{cases} x-b, & x \geq b \\ -(x-b), & b \geq x \end{cases}.$$

Continuing, we have

$$\begin{aligned}\langle \psi | \psi \rangle &= |A|^2 \left(e^{-2ab} \int_{-\infty}^b e^{2ax} dx + e^{2ab} \int_b^{\infty} e^{-2ax} dx \right) \\ &= |A|^2 \left(e^{-2ab} \frac{1}{2a} [e^{2ax}]_{-\infty}^b + e^{2ab} \frac{1}{-2a} [e^{-2ax}]_b^{\infty} \right) \\ &= \frac{|A|^2}{2a} \left(e^{-2ab} (e^{2ab} - 0) - e^{2ab} (0 - e^{-2ab}) \right) \\ &= \frac{|A|^2}{2a} (e^0 + e^0) \\ &= \frac{|A|^2}{a} \\ &\equiv 1.\end{aligned}$$

It follows that $A = \pm\sqrt{a}$.

2.

$$\begin{aligned}
\langle \psi | P | \psi \rangle &= \int_{-\infty}^{\infty} \psi^*(x) \left(-i\hbar \frac{d}{dx} \right) \psi(x) dx \\
&= a \int_{-\infty}^{\infty} e^{-a|x-b|-ivx} \left(-i\hbar \frac{d}{dx} \right) e^{-a|x-b|+ivx} dx \\
&= -i\hbar a \int_{-\infty}^{\infty} \left(e^{-a|x-b|-ivx} \right) \left(e^{-a|x-b|+ivx} \right) (-a \operatorname{sign}(x-b) + iv) dx \\
&= -i\hbar a \int_{-\infty}^{\infty} \left(e^{-2a|x-b|} \right) (-a \operatorname{sign}(x-b) + iv) dx \\
&= -i\hbar a \left(\int_{-\infty}^b \left(e^{2a(x-b)} (a + iv) \right) dx + \int_b^{\infty} \left(e^{-2a(x-b)} (-a + iv) \right) dx \right) \\
&= -i\hbar a (a + iv) \int_{-\infty}^b e^{2a(x-b)} dx - i\hbar a (-a + iv) \int_b^{\infty} e^{-2a(x-b)} dx \\
&= \frac{-i\hbar a^2 + \hbar a v}{2a} \left[e^{2a(x-b)} \right]_{-\infty}^b + \frac{i\hbar a^2 + \hbar a v}{-2a} \left[e^{-2a(x-b)} \right]_b^{\infty} \\
&= \frac{-i\hbar a + \hbar v}{2} (1 - 0) - \frac{i\hbar a + \hbar v}{2} (0 - 1) \\
&= \frac{-i\hbar a + \hbar v + i\hbar a + \hbar v}{2} \\
&= \hbar v.
\end{aligned}$$

$$\frac{d^2}{dx^2} \psi(x) = \left(e^{-a|x-b|+ivx} \right) (-a \operatorname{sign}(x-b) + iv)^2 + \left(e^{-a|x-b|+ivx} \right) (-2a\delta(x-b)).$$

We then obtain two integrals for $\langle \psi | P^2 | \psi \rangle$, corresponding to the two terms in the derivative. I will compute them in order, and add them together at the end.

$$\begin{aligned}
\text{Int 1} &= -a\hbar^2 \int_{-\infty}^{\infty} \left(e^{-a|x-b|-ivx} \right) \left(e^{-a|x-b|+ivx} \right) (-a \operatorname{sign}(x-b) + iv)^2 dx \\
&= -a\hbar^2 \int_{-\infty}^{\infty} \left(e^{-2a|x-b|} \right) (a^2 \operatorname{sign}(x-b)^2 - 2aiv \operatorname{sign}(x-b) - v^2) dx \\
&= -a\hbar^2 \int_{-\infty}^b \left(e^{2a(x-b)} \right) (a^2 + 2aiv - v^2) dx - a\hbar^2 \int_b^{\infty} \left(e^{-2a(x-b)} \right) (a^2 - 2aiv - v^2) dx \\
&= -a\hbar^2 (a^2 + 2aiv - v^2) \left[\frac{1}{2a} e^{2a(x-b)} \right]_{-\infty}^b + a\hbar^2 (a^2 - 2aiv - v^2) \left[\frac{1}{2a} e^{-2a(x-b)} \right]_b^{\infty} \\
&= -\frac{\hbar^2}{2} (a^2 + a^2 + 2aiv - 2aiv - v^2 - v^2) \\
&= \hbar^2 (v^2 - a^2).
\end{aligned}$$

For the second integral:

$$\begin{aligned}
\text{Int 2} &= -a\hbar^2 \int_{-\infty}^{\infty} \left(e^{-a|x-b|-ivx} \right) \left(e^{-a|x-b|+ivx} \right) (-2a\delta(x-b)) dx \\
&= a^2\hbar^2 \int_{-\infty}^{\infty} e^{-2a|x-b|} 2\delta(x-b) dx \\
&= 2a^2\hbar^2 e^{-2a|b-b|} = 2a^2\hbar^2.
\end{aligned}$$

Adding these together, we get

$$\langle \psi | P^2 | \psi \rangle = v^2 \hbar^2 - a^2 \hbar^2 + 2a^2 \hbar^2 = \hbar^2 (v^2 + a^2) .$$

3.

$$\begin{aligned} \langle \psi | X | \psi \rangle &= \langle \psi | \int_{-\infty}^{\infty} x | x \rangle \langle x | dx | \psi \rangle \\ &= \int_{-\infty}^{\infty} \langle \psi | x \rangle x \langle x | \psi \rangle dx \\ &= \int_{-\infty}^{\infty} x \psi^*(x) \psi(x) dx \\ &= a \int_{-\infty}^{\infty} x e^{-2a|x-b|} dx. \end{aligned}$$

Let $u = x - b$. Then $x = u + b$. I will also later do the substitution $u = -u$, where I do not change the name of the variable because I simply do not want to.

$$\begin{aligned} a \int_{-\infty}^{\infty} x e^{-2a|x-b|} dx &= a \int_{-\infty}^{\infty} (u + b) e^{-2a|u|} du \\ &= a \int_{-\infty}^{\infty} u e^{-2a|u|} du + ab \int_{-\infty}^{\infty} e^{-2a|u|} du \\ &= a \int_{-\infty}^{\infty} u e^{-2a|u|} du + b \int_{-\infty}^{\infty} |\psi(x)|^2 dx \\ &= a \int_{-\infty}^{\infty} u e^{-2a|u|} du + b \\ &= a \int_{-\infty}^0 u e^{2au} du + a \int_0^{\infty} u e^{-2au} du + b \\ &= a \int_0^{\infty} -(-u) e^{2a(-u)} du + a \int_0^{\infty} u e^{-2au} du + b \\ &= -a \int_0^{\infty} u e^{-2au} du + a \int_0^{\infty} u e^{-2au} du + b \\ &= 0 + b \\ &= b. \end{aligned}$$

$$\begin{aligned}
\langle \psi | X^2 | \psi \rangle &= a \int_{-\infty}^{\infty} x^2 e^{-2a|x-b|} dx \\
&= a \int_{-\infty}^{\infty} (u+b)^2 e^{-2a|u|} du \\
&= a \left(b^2 \int_{-\infty}^{\infty} e^{-2a|x-b|} dx + 2b \int_{-\infty}^{\infty} u e^{-2a|u|} du + \int_{-\infty}^{\infty} u^2 e^{-2a|u|} du \right) \\
&= b^2 + a \int_{-\infty}^{\infty} u^2 e^{-2a|u|} du \\
&= b^2 + 2a \int_0^{\infty} u^2 e^{-2au} du \\
&= b^2 + 2a \left(\left[\frac{1}{-2a} u^2 e^{-2au} \right]_0^{\infty} - \int_0^{\infty} \frac{2}{-2a} u e^{-2au} du \right) \\
&= b^2 + 2 \left(\left[\frac{1}{-2a} u e^{-2au} \right]_0^{\infty} - \int_0^{\infty} \frac{1}{-2a} e^{-2au} du \right) \\
&= b^2 + \frac{1}{a} \int_0^{\infty} e^{-2au} du \\
&= b^2 + \frac{1}{-2a^2} [e^{-2au}]_0^{\infty} \\
&= b^2 - \frac{1}{2a^2} (0 - 1) \\
&= b^2 + \frac{1}{2a^2}.
\end{aligned}$$

4.

$$\begin{aligned}
\mathcal{P}_{(0,1)} &= \int_0^1 |\psi(x)|^2 dx \\
&= \int_0^1 |\pm \sqrt{a}|^2 |e^{-a|x-b|} e^{ivx}|^2 dx \\
&= a \int_0^1 |e^{-a|x-b|}|^2 dx \\
&= a \int_0^1 e^{-2a|x-b|} dx.
\end{aligned}$$

If $b \in (0, 1)$, then the integral decomposes into

$$\begin{aligned}
\mathcal{P}_{(0,1)} &= a \left(\int_0^b e^{2a(x-b)} dx + \int_b^1 e^{-2a(x-b)} dx \right) \\
&= a \left(e^{-2ab} \int_0^b e^{2ax} dx + e^{2ab} \int_b^1 e^{-2ax} dx \right) \\
&= a \left(\frac{e^{-2ab}}{2a} [e^{2ax}]_0^b + \frac{e^{2ab}}{-2a} [e^{-2ax}]_b^1 \right) \\
&= \frac{1}{2} \left(e^{-2ab} (e^{2ab} - 1) - e^{2ab} (e^{-2a} - e^{-2ab}) \right) \\
&= \frac{1}{2} (1 - e^{-2ab} - e^{2a(b-1)} + 1) \\
&= 1 - \frac{1}{2} (e^{-2ab} + e^{2a(b-1)}).
\end{aligned}$$

If $b \geq 1$, then we can just integrate directly:

$$\begin{aligned}
\mathcal{P}_{(0,1)} &= a \int_0^1 e^{2a(x-b)} dx \\
&= ae^{-2ab} \int_0^1 e^{2ax} dx \\
&= \frac{e^{-2ab}}{2} [e^{2ax}]_0^1 \\
&= \frac{e^{-2ab}}{2} (e^{2a} - 1) \\
&= \frac{1}{2} (e^{2a(1-b)} - e^{-2ab}).
\end{aligned}$$

2 Spectroscopy

I will use the approximation that $\hbar \approx 6.6 \times 10^{-16}$ eVs, because I don't want to write out a lot of decimal places because it looks less clean.

1. Our definition of frequency was $\omega_{21} = \frac{E_2 - E_1}{\hbar}$. The “first” frequency we can define, corresponding to $n = 1$, would be ω_{21} . We thus have

$$E_2 - E_1 = E_2 - 0 = \frac{\hbar}{1(1+1)} s^{-1} = \frac{\hbar}{2} s^{-1} \approx 3.3 \times 10^{-16} \text{ eV}.$$

2. It follows extremely easily from mathematical induction that $|E_n - E_{n-1}| = \hbar\omega_{n-1}$, and since

$$\frac{E_n - E_{n-1}}{\hbar} = \omega_{n-1} \implies E_n = \hbar\omega_{n-1} + E_{n-1}.$$

Since the base case E_2 is positive and $\hbar\omega_n$ is always positive, by induction the sequence $\{E_n\}$ is monotonically increasing. Therefore $E_n > E_{n-1}$, and $|E_n - E_{n-1}| = E_n - E_{n-1}$. I will spell out the next induction proof since it is the main part of the computation. Note that $E_2 = \hbar\omega_1 = \hbar \sum_{1 \leq i < 2} \omega_i$, and suppose $E_n = \hbar \sum_{1 \leq i < n} \omega_i$ for some n . Then

$$E_{n+1} = \hbar\omega_n + E_n = \hbar\omega_n + \hbar \sum_{i=1}^{n-1} \omega_i = \hbar \sum_{i=1}^n \omega_i.$$

By induction, the statement is proven. We thus have

$$\lim_{n \rightarrow \infty} E_n = \hbar \sum_{i=1}^{\infty} \frac{1}{n(n+1)}.$$

Note that

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1}{n(n+1)} - \frac{n}{n(n+1)} = \frac{1}{n(n+1)} = \omega_n.$$

Therefore,

$$\hbar \sum_{i=1}^{\infty} \omega_i = \hbar \sum_{i=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right).$$

It is immediately obvious this is a telescoping series, but I will prove this by induction once again. Note that

$$\sum_{i=1}^2 \omega_i = \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} = 1 - \frac{1}{2+1}.$$

Now suppose this statement holds for arbitrary n , and observe that

$$\sum_{i=1}^{n+1} \omega_i = \frac{1}{n+1} - \frac{1}{n+2} + \sum_{i=1}^n \omega_i = \frac{1}{n+1} - \frac{1}{n+2} + 1 - \frac{1}{n+1} = 1 - \frac{1}{n+2}.$$

Therefore, the statement is proven. We thus have

$$\hbar \sum_{i=1}^{\infty} \omega_i = \lim_{n \rightarrow \infty} \hbar \sum_{i=1}^n \frac{1}{n} - \frac{1}{n+1} = \hbar \lim_{n \rightarrow \infty} 1 - \frac{1}{n+1} = \hbar.$$

Therefore, and factoring in the s^{-1} that I have omitted writing for the entirety of the problem,

$$\lim_{n \rightarrow \infty} E_n = \hbar s^{-1} \approx 6.6 \cdot 10^{-16} \text{eV}.$$

3. (a) We have

$$\begin{aligned} \langle \psi | \psi \rangle &= A^2 \left(\sum_{n=1}^{\infty} z^n \langle E_n | \right) \left(\sum_{m=1}^{\infty} z^m | E_m \rangle \right) \\ &= A^2 \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} z^n z^m \delta_{nm} \\ &= A^2 \sum_{n=1}^{\infty} z^{2n} \\ &\equiv 1. \end{aligned}$$

Note that

$$\sum_{i=1}^{\infty} (z^2)^i$$

is geometric, and as $|z| < 1$, it converges to

$$\frac{z^2}{1 - z^2}.$$

We then define

$$A := \sqrt{\frac{1 - z^2}{z^2}}.$$

(b) i could have totally used einstein notation for this

$$\begin{aligned}
\langle \psi | H | \psi \rangle &= A^2 \left(\sum_{n=1}^{\infty} z^n \langle E_n | \right) \left(\sum_{m=1}^{\infty} z^m H | E_m \rangle \right) \\
&= A^2 \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} z^{nm} E_m \delta_{nm} \\
&= A^2 \sum_{n=1}^{\infty} (z^2)^n E_n \\
&= \hbar A^2 \sum_{n=1}^{\infty} (z^2)^n \left(1 - \frac{1}{n} \right) \\
&= \hbar A^2 \left(\frac{z^2}{1 - z^2} + \ln(1 - z^2) \right) \\
&\approx 6.6 \cdot 10^{-16} \cdot A^2 \left(\frac{z^2}{1 - z^2} + \ln(1 - z^2) \right) \text{ eV}.
\end{aligned}$$

I'm bored heres a proof of the identity

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \implies \int \frac{1}{1-x} dx = \sum_{n=0}^{\infty} \int x^n dx = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Also note that

$$\int \frac{1}{1-x} dx = -\ln(1-x).$$

Therefore,

$$-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n}.$$